Periodic Safety Update Report

Rotating and cutting instruments (connected to an active device) — extract: PMS information (b) to (d), MDCG 2022-21 Annex I

Cover page item	Entry
Manufacturer	Medartis AG (SRN CH-MF-000022291)
Medical device(s) covered	Rotating and cutting instruments (connected to an active device) — Basic UDI-DI 76300378015PD; 5 UDI-DIs; Class IIa; EMDN Z12130503 ORTHOPAEDIC SURGICAL DRILLS
Notified body	NB 0197 — certificate HZ 1809678-1
PSUR reference number / version	[manufacturer]
Data collection period	2024-10-07 to 2026-10-08

Post-Market Surveillance: information including general PMCF information

b) Scientific literature review of relevant specialist or technical literature

A literature search was performed on 2026-10-08 in PubMed/MEDLINE, Europe PMC and Crossref for records first available between 2024-10-07 and 2026-10-08, using a structured search for the device type, its synonyms and subject headings (full search strings in the search documentation). 1751 records were screened on title and abstract; 106 met the inclusion criteria (the device type is a subject of the publication and safety, performance or clinical-use findings are reported), of which 24 report safety-related findings. Publications concerning the device(s) covered by this PSUR: 0.

#	Publication	Finding type
1	Ou C et al. (2026). Ultrasonic Versus High-Speed Drilling for C2 Pedicle Screw Placement with High-Riding Vertebral Artery: A Comparative Study of Vascular Safety and Thermal Injury. Annals of biomedical engineering.	Safety
2	Tur D et al. (2026). Thermal evaluation of pilot- and partially guided implant surgery - an in vitro study using automated robotic drilling. BMC oral health.	Safety
3	Jahwari A et al. (2026). A Comparative Study of Piezoelectric Surgery versus High-Speed Drill in Cervical Spine Laminectomy. Journal of orthopaedic case reports.	Safety
4	Dalmage M et al. (2026). Drill choice as a modifiable factor in pediatric sEEG safety. Child's nervous system.	Safety
5	Balkanlioğlu AC et al. (2026). Effect of Osteotomy Depth on Intraosseous Temperature During Implant Drilling: A Standardized CAD/CAM-Guided Thermocouple Study. The International journal of oral & maxillofacial implants.	Safety
6	Zhu K et al. (2026). Force-sensing drill-through detection and automatic stopping in a porcine distal humerus model simulating pediatric supracondylar pinning. Frontiers in pediatrics.	Safety

Table continues: 100 further publications.

Manufacturer's assessment against the risk management file and clinical evaluation: [to be completed by the manufacturer].

c) Public databases and/or registry data

Database reviewed	Type	Period	Similar devices: findings
FDA MAUDE (USA)	Adverse event reports	2024-10-07 – 2026-10-08	5,032 reports (death 0, injury 117, malfunction 4,915); product code HTW, HWE, DZI; brands of this device type only
FDA Medical Device Recalls (USA)	Field safety corrective actions	5 years to 2026-10-08	2 recall events, 1 in the data collection period
Health Canada Recalls and Safety Alerts	Field safety corrective actions	5 years	1 recall
MHRA Field Safety Notices (UK)	Field safety notices	5 years	3 notices
EUDAMED (EU)	Device registration	at 2026-10-08	3,175 UDI-DIs registered under EMDN Z12130503

Most frequent device problems, similar devices (FDA MAUDE)	Reports
Break	1,877
Physical Resistance/Sticking	1,315
Complete Loss of Power	694
Mechanical Jam	387
Fracture	368

Most frequent health effects, similar devices (FDA MAUDE)	Reports
Foreign Body In Patient	397
Device Embedded In Tissue or Plaque	15
Perforation	6
Bone Notching	5

66% of the MAUDE reports for similar devices in the period were submitted by one manufacturer (Synthes); year-on-year changes mainly reflect that manufacturer's reporting. Reports concerning the device(s) covered by this PSUR are excluded from these figures.

Date	Firm / authority	Field action (excerpt)
2026-02-11	Waldemar Link GmbH & Co. KG (Mfg Site)	The surgical technique for the Embrace Shoulder Instruments - Drill Tower has mixed-up article numbers in the overview of the instruments. However, the …
2024-01-29	Waldemar Link GmbH & Co. KG (Mfg Site)	The surgical techniques for the Embrace Shoulder Instruments -Drill Tower (Article REF 645-081/62 & /63) have mixed-up type descriptions ("standard" vs …
2024-11-04	Health Canada	JuggerLoc Disposable Cannula Kit - Drill Guide
2026-07-14	Arthrex	3.4 mm Drill Bit for SwiveLock 4.75 mm, Reusable
2024-11-12	Zimmer Biomet	JuggerLoc™ Disposable Cannula Kit - Drill Guide

Registries: no device registry with public device-level data was identified for this device type.

Manufacturer's assessment against the risk management file and clinical evaluation: [to be completed by the manufacturer].

d) Publicly available information about similar medical devices

Similar devices are those registered under EMDN Z12130503 (ORTHOPAEDIC SURGICAL DRILLS) and, for third-country data, FDA product code HTW, HWE, DZI. In EUDAMED, 13 other manufacturers were identified in a sample of 208 registrations. In the USA, no new similar devices received 510(k) clearance in the period.

Manufacturer of similar devices (EU market)	Country
WILSON (Shanghai) Medical Device Co., Ltd.	China
Synthes GmbH	Switzerland
Aesculap AG	Germany
Conmed Corporation	United States

Summary of safety and clinical performance (SS(C)P): not applicable — the device is Class IIa and non-implantable (MDR Art. 32).

Manufacturer's assessment against the risk management file and clinical evaluation: [to be completed by the manufacturer].